

ServiceNow Hackathon Build Guide

HackNow India 2026 (ServiceNow University) — Zero to Hero Development Plan

Patient Discharge Readiness Command Center

Team Ignite — Mohan Babu University — Problem Statement #HC-01

This guide is a complete, step-by-step build plan written in plain language: exact tables, fields, roles, flows, SLA configuration, dashboard, portal pages, demo data, and a demo script — all built directly from Team Ignite's submitted Idea Submission PDF, so what you configure in the instance matches exactly what was pitched.

Table of Contents

Step 0 — What We're Building (In Plain Words)

Step 1 — Create the Scoped Application

Step 2 — Create the Tables (6 tables)

Step 3 — Roles & Access Control

Step 4 — Build the Flows (4 flows, matching the submitted workflow diagram)

Step 5 — The Real-Time Command Center Dashboard

Step 6 — Build the Portal / Workspace

Step 7 — Seed Demo Data

Step 8 — Demo Script

Patient Discharge Readiness Command Center

Step 0 — What Are We Actually Building? (In Plain Words)

Picture a patient in a hospital bed. The doctor says “medically fine to go home” — but the patient doesn't actually leave for hours or even days, because:

- Billing hasn't finished the bill
- The insurance company (TPA) hasn't approved the claim
- Pharmacy hasn't packed the discharge medicines
- Transport hasn't been arranged
- Nobody has scheduled the follow-up visit

Right now, all of this is tracked in people's heads, phone calls, and separate spreadsheets. Nobody can answer “why is this specific patient still here?” in one glance.

We're building a system that, the moment a doctor says “ready,” automatically:

- Creates one central case for that patient's discharge
- Splits it into 6 tasks (one per department) and sends each to the right team
- Calculates a **percentage score** showing how close the patient is to actually leaving
- Watches the clock on every task, and if one is about to run late, warns people **before** it's late
- If a task keeps getting bounced between people (reassigned again and again), it flags that as a warning sign on its own
- Escalates to a supervisor automatically if something is stuck
- Shows managers one big screen with every blocked patient, every at-risk deadline, and how many hospital beds are stuck because of it

In ServiceNow, everything lives in a **table** (like an Excel sheet — rows and columns) and things happen automatically because of **flows** (if-this-happens-then-do-that recipes) and **SLAs** (built-in ServiceNow stopwatches that track deadlines).

Step 1 — Create Your App's Home (Scoped Application)

Why: Everything you build needs its own named “folder” in ServiceNow so it doesn't get mixed up with anything else on the instance.

How:

- 1 In the top search bar (the **Application Navigator**), type Studio, open it
- 2 Click **Create Application** → **Create Application from Scratch**
- 3 Name it: **Discharge Command Center**
- 4 Scope auto-fills as x_yourprefix_discharge_cc — leave it
- 5 Click **Create**

Everything below lives inside this container.

Step 2 — Create Your Tables (the Excel-Sheet Part)

We need **6 tables**. For each: Create Application File → Table → give it a Label → add fields.

Table 1: Patient

One row per patient currently admitted.

Field Label	Field Type	Notes
Patient Name	String	
Ward / Bed Reference	Reference (to Table 6, Bed)	Which bed they're in
Attending Physician	Reference (to sys_user)	The doctor
Medically Ready	True/False (checkbox)	The doctor flips this switch
Medically Ready Date/Time	Date/Time	Stamped the moment the checkbox flips
Discharge Readiness Score	Integer (%)	Starts blank, filled in by a flow
Status	Choice	Admitted / Discharge In Progress / Discharged

Baby explanation: Think of this like the whiteboard nurses keep at the nurses' station with each patient's name and status — except now it updates itself.

Table 2: Discharge Case

The one central “folder” created the moment a patient becomes medically ready.

Field Label	Field Type	Notes
Patient	Reference (to Table 1)	
Case Number	Auto-generated (Number field)	ServiceNow can auto-number this for you
Status	Choice	Open / At Risk / Blocked / Ready for Discharge / Closed
Expected Discharge Date/Time	Date/Time	Set when the case is created
Overall Readiness %	Integer	Rolls up from the 6 tasks underneath it
Has Active Blocker	True/False	Flips on if any task is stuck

Baby explanation: This is the single “cover page” of a folder — everything about this patient's discharge is clipped underneath it, so nobody has to go hunting across six different departments' notebooks.

Table 3: Discharge Task

The 6 individual jobs — Documentation, Billing, TPA/Insurance, Pharmacy, Transport, Follow-up — that get created automatically for every case.

Field Label	Field Type	Notes
Discharge Case	Reference (to Table 2)	
Task Type	Choice	Documentation / Billing / TPA-Insurance / Pharmacy / Transport / Follow-up
Assigned Group	Reference (to sys_user_group, native)	Which department team owns it
Assigned To	Reference (to sys_user)	The specific person, once picked up
Status	Choice	Open / In Progress / Blocked / Completed
Priority	Choice	Low / Medium / High / Critical
Reassignment Count	Integer	Starts at 0, +1 every time it's reassigned
Blocker Notes	String (multi-line)	Filled in if someone marks it Blocked

Baby explanation: Why not just one big task? Because six different teams (billing, pharmacy, transport...) each need their own to-do item, and each one needs its own clock ticking separately — otherwise you can't tell which department is the one holding everyone up.

This table is also where ServiceNow's native SLA feature attaches — see Step 4, Flow 3.

Table 4: Escalation Log

A record every time something gets kicked upstairs.

Field Label	Field Type	Notes
Discharge Task	Reference (to Table 3)	
Escalation Reason	Choice	SLA Breach Risk / SLA Breached / Repeated Reassignment / Manual Escalation
Escalated To	Reference (to sys_user)	The department lead or coordinator
Escalated Date/Time	Date/Time	
Resolved Date/Time	Date/Time	Blank until closed
Status	Choice	Open / Resolved

Baby explanation: This is like a “we need a grown-up” log — every time a task can't get itself sorted out in time, this table writes down exactly what happened and who got called in to fix it.

Table 5: Department Team

Shortcut — don't build this as a custom table. ServiceNow already has native **Groups** (sys_user_group) built in. Just create 5 groups: Billing Team, TPA/Insurance Team, Pharmacy Team, Transport Team,

Documentation Team — with the right staff added as members. This saves you a whole table and shows you know when to use what's already there instead of rebuilding it.

Table 6: Bed

Tracks which beds are occupied vs. free — this is what makes the “Bed-Day Impact Visibility” feature from your submission possible.

Field Label	Field Type	Notes
Bed Number	String	e.g. “Ward 3 - Bed 12”
Ward	Choice or Reference	
Status	Choice	Occupied / Available / Cleaning
Current Patient	Reference (to Table 1)	Blank when the bed is free

Step 3 — Roles (Who's Allowed to Do What)

Baby explanation: A role is a wristband — it decides which doors you can walk through.

Create these **5 roles** (Studio → Create Application File → Role):

- 1 **Physician** — can flip the “Medically Ready” switch on a Patient record
- 2 **Department Staff** — can see and update tasks assigned to their own group (billing sees only billing tasks, pharmacy sees only pharmacy tasks, etc.)
- 3 **Discharge Coordinator** — can see every case, every task, resolve blockers, mark cases ready
- 4 **Department Lead / Supervisor** — receives escalations for their department, can reassign tasks
- 5 **Admin** — full access, this is you the developer

Then set ACLs, same pattern as always: a Department Staff ACL on Discharge Task should only allow read/write where “Assigned Group = my group” — otherwise the pharmacy team could see and edit billing's tasks, which you don't want.

Step 4 — Build the Flows (Matching Your Own “How It Works” Diagram)

Your submission already drew the exact 8-step logic you need — we're just going to build it precisely as diagrammed, split into 4 flows.

Flow 1: Case Creation & Parallel Task Generation

Trigger: Record Updated → Table = Patient → Condition: Medically Ready changes to **true**

- 1 **Create Record** → Table = Discharge Case → Patient = this patient, Status = “Open,” Expected Discharge Date/Time = now + 24 hours (or whatever your team decides as a target)

- 2 **Create Record** (repeat 6 times, once per Task Type) → Table = Discharge Task → Discharge Case = the case just created, Task Type = Documentation / Billing / TPA-Insurance / Pharmacy / Transport / Follow-up, Assigned Group = the matching department group, Status = "Open"
- 3 **Send Notification** → to each Assigned Group → "New discharge task ready for [Patient Name]"

Baby explanation: The moment the doctor flips one switch, six invisible "to-do" notes fly out to six different teams at the exact same time — nobody has to remember to tell anyone.

Flow 2: Discharge Readiness Score ("Patient-Centric Readiness Score")

Trigger: Record Updated → Table = Discharge Task → Condition: Status changes to any value

- 1 **Look Up Records** → all Discharge Tasks where Discharge Case = this task's case
- 2 **Count** how many are "Completed" out of the total 6
- 3 **Math step:** Overall Readiness % = (Completed Count ÷ 6) × 100
- 4 **Update Record** → Discharge Case → set Overall Readiness %
- 5 **If/Else:** if any task in that lookup has Status = "Blocked" → set Discharge Case's Has Active Blocker = true; otherwise false

Baby explanation: Imagine a progress bar on a phone download — every time one of the six tasks finishes, the bar fills in a little more, so anyone glancing at the case instantly knows "this one's 66% done," instead of having to check six separate to-do lists.

Flow 3: Predictive SLA Warning & Escalation

This is where you use ServiceNow's **native SLA Management** module — you don't have to build the clock yourself, ServiceNow already has one.

Setup (not a flow, a configuration):

- 1 Type SLA Definitions in the navigator → New
- 2 Table = Discharge Task, Start condition = Status is "Open," Stop condition = Status is "Completed"
- 3 Set the duration (e.g. 4 hours for Pharmacy, 6 hours for Billing — different SLA definitions per Task Type if you want different urgency per department)
- 4 Turn on **Workflow** for warning percentage — ServiceNow can automatically mark a task "breached" or "at risk" at, say, 80% of the time used

Then build the flow on top of it:

Trigger: Record Updated → Table = Discharge Task → Condition: SLA stage changes to "At Risk" OR "Breached"

- 1 **If/Else:** stage = "At Risk" → **Send Notification** → to Assigned Group → "Warning: this task is close to its deadline"
- 2 **If/Else:** stage = "Breached" → **Create Record** → Table = Escalation Log → Reason = "SLA Breached" → Escalated To = the Department Lead for that group → **Send Notification** to that lead

Separate small flow for the Reassignment Trigger:

Trigger: Record Updated → Table = Discharge Task → Condition: Assigned To changes

- 1 **Update Record** → Reassignment Count = Reassignment Count + 1
- 2 **If/Else:** Reassignment Count ≥ 3 → **Create Record** → Escalation Log → Reason = “Repeated Reassignment” → notify the Department Lead

Baby explanation: This is like a teacher who doesn't just yell at you the moment homework is late — she warns you the day before it's due, and if you keep handing your homework to different classmates instead of doing it yourself, she notices that pattern too and steps in.

Flow 4: Discharge Completion & Bed Release

Trigger: Record Updated → Table = Discharge Task → Condition: Status changes to “Completed”

- 1 **Look Up Records** → all 6 Discharge Tasks for this Discharge Case
- 2 **If/Else:** are ALL 6 now “Completed”? If yes: **Update Record** → Discharge Case → Status = “Ready for Discharge” → **Send Notification** → to the Physician and Discharge Coordinator
- 3 **(Separate step, triggered when Coordinator marks the Patient as actually Discharged): Update Record** → Patient → Status = “Discharged” → **Update Record** → Bed → Status = “Available,” Current Patient = blank

Baby explanation: This is the finish line — the instant every single one of the six little jobs is checked off, the system automatically tells the doctor “this patient is genuinely ready to walk out,” and once they leave, the bed flips back to “open for the next patient” without anyone manually updating a spreadsheet.

Step 5 — The Real-Time Command Center Dashboard

Baby explanation: This is the “big screen” a hospital manager looks at to see the whole floor at once, instead of walking room to room.

Build this using **Performance Analytics** or plain **Reports** (Reports is the quicker, lighter option if you're short on time):

- **Medically Ready Patients** — count of Patient records where Medically Ready = true and Status ≠ Discharged
- **Blocked Cases** — count of Discharge Case where Has Active Blocker = true
- **SLA-at-Risk / Breached Tasks** — count of Discharge Task grouped by SLA stage
- **Pending Tasks by Department** — bar chart of open Discharge Tasks grouped by Task Type
- **Discharge Delay Heatmap** — a report grouped by Task Type showing which department most often causes delays (count of “Breached” SLA tasks per Task Type — the type with the most breaches is your bottleneck)
- **Bed Availability Impact** — count of Beds where Status = “Occupied” AND the linked Patient's Discharge Case Status = “Ready for Discharge” (this shows exactly how many beds are stuck even though the patient could leave)

Step 6 — Build the Portal / Workspace

Your submission lists **ServiceNow Workspace**, **Service Portal / UI Builder** — for an intermediate team, Service Portal is the faster, safer choice over building a full custom Workspace.

Pages to build:

- **Doctor view:** a simple list of their patients with a one-click “Mark Medically Ready” button
- **Department Staff view:** list of tasks assigned to their group only, with Status dropdown and a “Mark Blocked + add note” option
- **Coordinator/Command Center view:** the full dashboard from Step 5, plus a list of all active Discharge Cases sorted by Readiness %

Step 7 — Seed Demo Data

Before demo day, create:

- ~10-15 Patient records across a few wards, some already Medically Ready
- For 5-6 of them, let Flow 1 auto-generate their Discharge Cases and Tasks
- Manually complete some tasks, leave a few “Blocked” with notes, and deliberately let one or two run past their SLA so Flow 3’s escalation has something real to show
- A few Beds marked Occupied, one or two linked to a patient whose case is already “Ready for Discharge” — this is exactly what “Bed-Day Impact” needs to show

Step 8 — Your Demo Script (Matches Your Own Workflow Diagram)

- 1 Show a Patient record → doctor flips “Medically Ready” → watch the Discharge Case and all 6 tasks get created live
- 2 Complete a couple of tasks on camera → show the Readiness % climb in real time
- 3 Mark one task “Blocked” with a note → show it flip the case to “Has Active Blocker” and appear on the dashboard
- 4 Show a task that’s deliberately past its SLA → show the auto-created Escalation Log entry and the notification that went to the department lead
- 5 Reassign a task 3 times quickly → show the Smart Reassignment Trigger fire its own escalation
- 6 Complete the last task → show the case flip to “Ready for Discharge” live
- 7 Mark the patient discharged → show the Bed flip back to “Available”
- 8 Finish on the Command Center dashboard — readiness scores, blocked cases, the delay heatmap, and the bed-impact number all update together

Good luck, Team Ignite — build Flow 1 and Flow 2 first (case creation + readiness score), since everything else layers on top of those two. Seed realistic demo data early, and rehearse the demo script out loud before presenting.